

15/10/2025

Task-10

Use Matplotlib Module for plotting in python.

10.1

Aim: To use Matplotlib Module for plotting in python.

Algorithm:

- 1) Define two lists for programming languages and their popularity respectively.
- 2) Find the maximum popularity value in the list.
- 3) Define a scaling factor to scale the bar heights within a certain limit.
- 4) For each language and popularity pair, calculate the bar height as the popularity.
- 5) print the chart using a loop to iterate over the programming language list:
 - a) print the language name
 - b) use a loop to print
 - c) print the popularity.

Program:

```
#pip instal matplotlib
```

```
import matplotlib.pyplot as plt
```

```
languages = ['Java', 'python', 'PHP', 'Javascript', 'C#',  
             'C++']
```

```
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
```

```
plt.bar(languages, popularity, color='b')
```

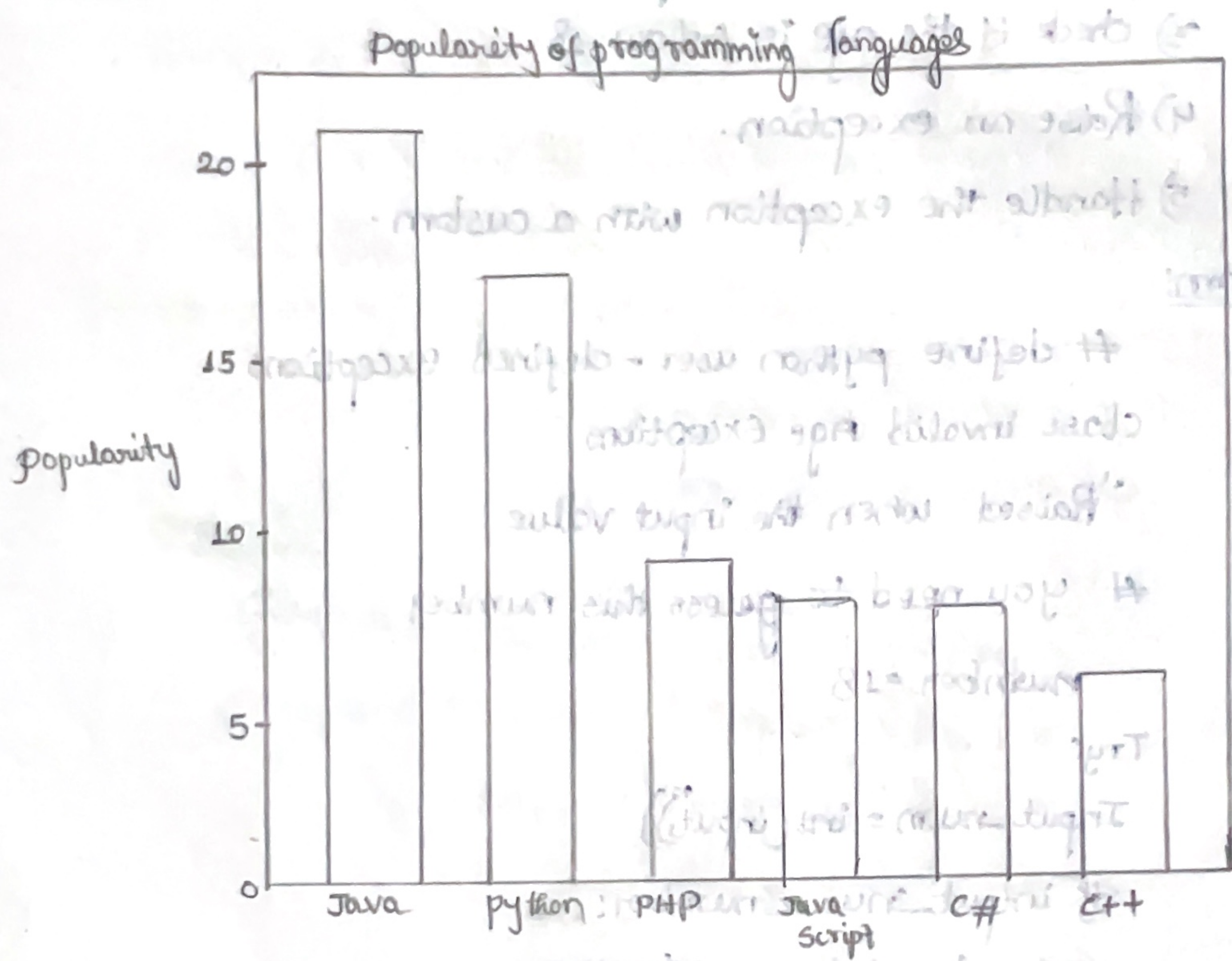
```
plt.title('popularity of programming languages')
```

```
plt.xlabel('programming languages')
```

```
plt.ylabel('popularity')
```

```
plt.show()
```


output

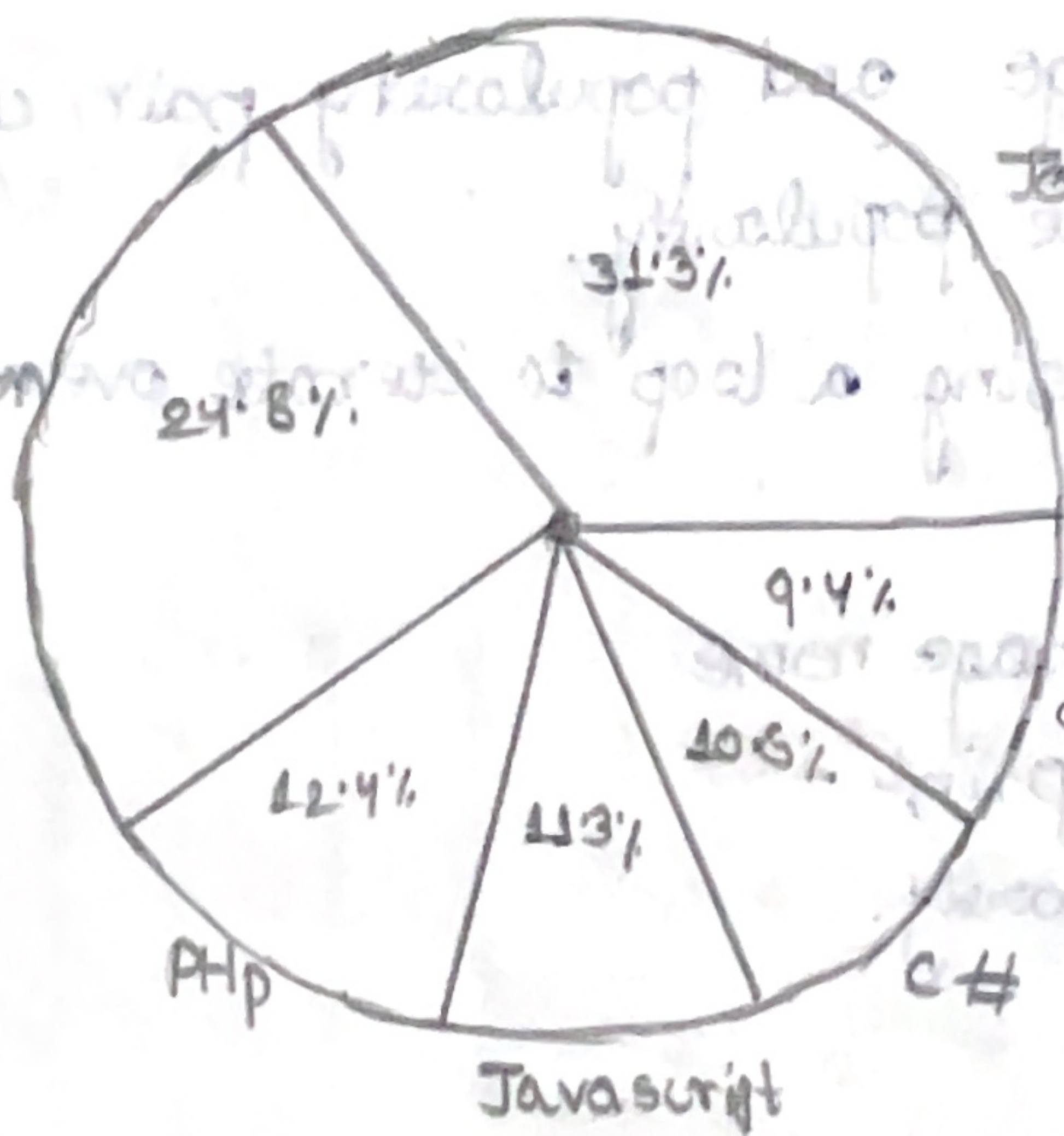


Results: True the program for implement a class and
Exceptional handling is executed and verified

Successfully

output:

popularity of programming languages



Task-10.2

→ write a python programming to create a pie chart of the popularity of programming languages

Algorithm:

- 1) Create a list of programming languages and popularity
- 2) Create a pie chart using the matplotlib library
- 3) Set the title and legend for the pie chart.
- 4) Show the pie chart.

Program:

```
import matplotlib.pyplot as plt
```

```
#step 1
```

```
languages = ['Java', 'python', 'php', 'Javascript', 'c#', 'c++']
```

```
popularity = [22.2, 47.6, 8.8, 8, 7.7, 6.7]
```

```
#step 2
```

```
plt.pie(popularity, labels=languages, autopct='%1.1f%%')
```

```
#step 3
```

```
plt.title('popularity of programming languages')
```

```
plt.legend(languages, loc="best")
```

```
#step 4
```

```
plt.show()
```

VEL TECH	
EX NO.	10
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	15

Result: Thus the python program use matplotlib module for plotting is executed and verified successfully.